# Assignment 6

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## Setup

I have mainly followed the tutorial available at https://docs.dbos.dev/python/examples/customer-service

I initialize my local directory and installed the necessary libraries

```
uv init --bare
uv add dbos
uv add fastapi
```

I have later decided to use Python 3.14, so I ran `uv python pin 3.14`

I created a `setup.sh` to collect and setup all the necessary environment variables like the AI key.

I copied the `html` folder from the original git repository so that I could create the database.

## Conversion to Gemini and Sqlite

Because of personal preferences, I have decided to convert the project to use sqlite and Google Gemini.

I have copied and pasted my Gemini key in the `setup.py` and installed some additional packages

- `uv add langgraph-checkpoint-sqlite`
- `uv add langchain-google-genai`

My local Gemini was able to help me convert the project to use sqlite and langchain-google-genai (the full conversation is available)

Gemini was able to convert the db.py (which was the reset_database.py to use sqlite.

I have also added the ability to start the server by running the `app.py` directly.

## Testing

I was able to create the database and start the FastAPI server.

```
source setup.sh
uv run db.py
uv run app.py
```

After the initial chat, I was receiving messages from Gemini and some of the messages got corrupted due to the incorrect decoding by the api endpoint.

```
Gemini: I can help you look up your orders, but I'll need an order ID. Do you have that hand
User: No I don't know them
Gemini: [object Object]
```

So Gemini was able to fix it by rendering correctly the dictionary response from Gemini, probably because it was not accounted for.

I thought it was very curious to see the Gemini response when it was not parsed correctly.

```
[{'type': 'text', 'text': "Okay, I'll explain it again, simply.\n\nMy main functions are:\n\
```

The DBOS Customer service finally worked correctly.

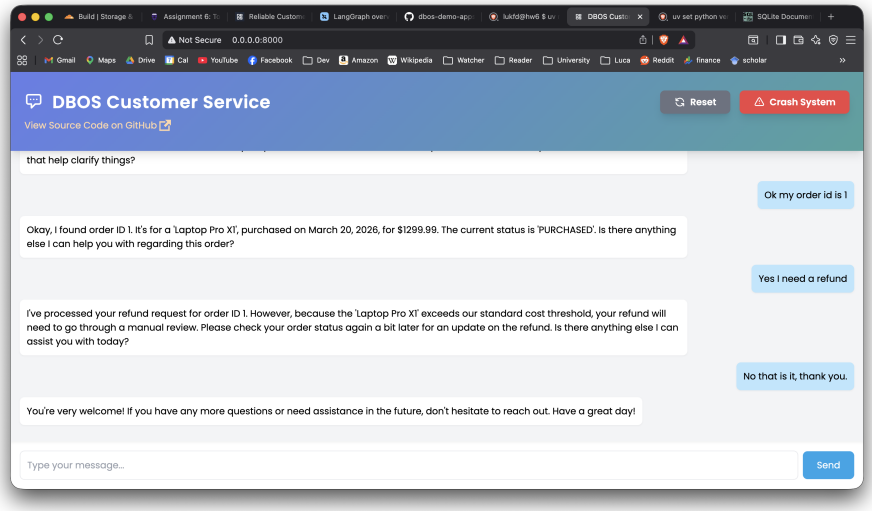


Figure 1: image 1

## Gemini Help

I have exported the entire conversation I had with Google Gemini as it helped me debug the code and better understand the code base.